

Quality improvement initiative: implementing and redefining video review of real-time neonatal procedures using action research

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ABSTRACT

Video review (VR) of procedures in the medical environment can be used to drive quality improvement. However, first it has to be implemented in a safe and effective way. Our primary objective was to (re)define a guideline for implementing interprofessional VR in a neonatal intensive care unit (NICU). Our secondary objective was to determine the rate of acceptance by providers attending VR. For 9 months, VR sessions were evaluated with a study group, consisting of different stakeholders. A questionnaire was embedded at the end of each session to obtain feedback from providers on the session and on the safe learning environment. In consensus meetings, success factors and preconditions were identified and divided into different factors that influenced the rate of adoption of VR. The number of providers who recorded procedures and attended VR sessions was determined. A total of 18 VR sessions could be organised, with an equal distribution of medical and nursing staff. After the 9-month period, 101/125 (81%) of all providers working on the NICU attended at least 1 session and 80/125 (64%) of all providers recorded their performance of a procedure at least 1 time. In total, 179/297 (61%) providers completed the questionnaire. Almost all providers (99%) reported to have a positive opinion about the review sessions. Preconditions and success factors related to implementation were identified and addressed, including improving the pathway for obtaining consent, preparation of VR, defining the role of the chair during the session and building a safe learning environment. Different strategies were developed to ensure findings from sessions were used for quality improvement. VR was successfully implemented on our NICU and we redefined our guideline with various preconditions and success factors. The adjusted guideline can be helpful for implementation of VR in emergency care settings.

PROBLEM

Providing optimal care in a neonatal intensive care unit (NICU) is important for short-term and long-term outcomes of infants born premature or who are severely ill.^{1 2} However, this can be difficult as the NICU is characterised by a dynamic, multidisciplinary and emergent environment.³ Guidelines

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Implementation of regularly organised video review sessions using real-time recorded procedures offers many possibilities for improving quality of care. However, organisation of interprofessional video review while maintaining a safe learning environment in emergency or intensive care can be difficult and a guideline is lacking.

WHAT THIS STUDY ADDS

⇒ This study describes an initiative to improve neonatal care by implementing video review. Video review can be successfully implemented using an action research approach. Success factors and preconditions for implementation have been identified. Providers were positive and accepted the implementation of video review as standard care.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Our redefined guideline with various preconditions and success factors provides direction for future use of video review and for implementation of video review in other emergency or intensive care settings.

are created to offer guidance, but there is still considerable variation in care.^{4–6} Consequently, the variability suggests that there is opportunity to learn from each other and to improve care. Video recording of real-time performed procedures provides an objective view and can be used for improving complex procedures.^{7–14}

Recording video and physiological parameters of neonatal resuscitation at birth have been used for research in the NICU of the Leiden University Medical Center (LUMC) since 2009, capturing only imaging of the neonate and providers' hands. The LUMC is a tertiary-level hospital with a 25-bed level-III NICU, annually admitting approximately 650 inborn and outborn patients. The ward consists of 16 single-patient rooms and 4 twin rooms. On average, 125 providers work on

the NICU who are involved in hands-on care, including 95 nursing staff members and 30 medical staff members. From 2014 onwards, the recordings of neonatal stabilisation were reflected on weekly, traditionally with mostly medical staff. This improved guideline compliance and documentation.^{5 15–17} Neonatal care providers viewed video review (VR) of neonatal stabilisation as beneficial, but emphasised that a safe learning environment was essential. Members of the medical and nursing staff expressed an interest in expanding VR with audio to evaluate non-technical aspects as well, such as communication and teamwork.¹⁸ Additionally, staff members wished for all providers to be able to participate in VR, including nursing staff. Finally, providers took an interest in recording medical procedures in the NICU environment as well, next to recording procedures in the delivery room.

Implementation of VR to (1) include audio, (2) record procedures in the NICU environment as well as in the delivery room, (3) involve not only the medical staff but also the nursing staff in VR sessions and (4) strive for a safe learning environment during the sessions is a relatively unexplored area in the use of VR.^{7 18–21} The primary aim of this study was to evaluate how to successfully use VR in our NICU. We aimed to organise 18 VR sessions in 9 months evaluating real-time audio–visual recordings of neonatal procedures. Procedures had to be recorded on a voluntary basis, by different providers of the NICU team. During each session, we aimed to have an equal distribution of members from the medical and nursing staff. We were also interested in whether providers attending VR regarded the sessions as positive and felt that a safe learning environment was present. During the implementation of VR, we (re)defined our guideline for VR using continuous improvement cycles.

BACKGROUND

VR has proven to be beneficial for guideline compliance and clinical performance of emergency and intensive care teams.^{7 12 17 22} Although use of VR is becoming more accepted in healthcare, using real-time recordings of clinical procedures is not standardly used as a tool for improving care.⁷ When videos are used for VR with all providers present, it can give feelings of distress for being recorded.²³ Providers can fear criticism or judgement from colleagues or the risk of being seen as less competent.^{24 25} Next to that, VR can be time consuming.^{26 27} Studies on VR are often not focused on reporting how to set up regularly organised, interprofessional VR in a way that is safe and suitable for daily practice and can be embedded on the long term.^{27–29} There were no practical guidelines available for the implementation of this particular application of VR into the daily practice in an intensive care setting. We started the design of a new, expanded VR program, and named this the *Neoflix* project. Increasing the understanding of what is needed to successfully implement and use VR in the daily routine

of emergency or intensive care departments might help other departments to make use of the potential VR has to offer.

MEASUREMENT

A previous study in our unit on the use of VR,¹⁵ conducted between February 2018 and February 2019, was used as a baseline indicator. This study showed that 3 VR sessions were organised each month, with on average 14 medical staff members and 1 nursing staff member attending. The procedures that were recorded included neonatal stabilisation only. Imaging of the hands of the providers and the neonate were captured without audio, resulting in unidentifiable videos.

This study used routinely collected data on VR from December 2021 to September 2022. Procedures that could be (audio-visually) recorded included neonatal stabilisation, sterile line insertions and endotracheal intubation/surfactant administration. Outcome measures included the number of providers who recorded any of these procedures, how many recordings were made per procedure and the number of providers attending the VR sessions. Other outcome measures included the opinion of the providers on the VR sessions and on the safe learning environment.

DESIGN

We considered that the implementation of VR needed to be organised and adjusted step-by-step along the way, together with the providers who deal with VR in practice. Therefore, this study was designed, using an action research approach, to ensure implementation was provider driven. First, a *Neoflix* study team was formed, consisting of members from the senior and junior medical staff, nursing staff, a physician assistant and a member from the quality and patient safety department. This group was composed to represent all different stakeholders involved in neonatal care. A predefined guideline was designed for implementation of VR (online supplemental file 1), including which procedures were to be recorded on a voluntary basis (ie, neonatal stabilisation at birth, intubation and surfactant administration and sterile line insertions) (figure 1), the consent procedure for providers and patients, and details on the organisation of VR in general. We aimed to use the providers' perspective to drive continuous improvement cycles during implementation to improve and adapt the guideline after each VR session to overcome challenges that influenced the rate of adoption and acceptance. The *Neoflix* study group met every 2 weeks to monitor the implementation and make adjustments.

STRATEGY

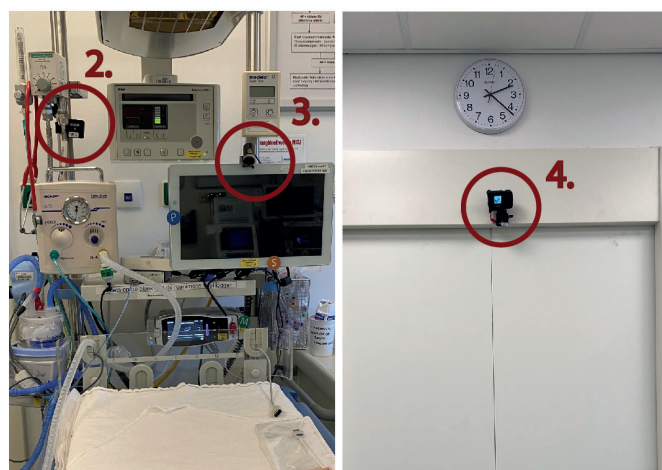
From December 2021 until September 2022, the preparation, the organisation and the conduct of the VR sessions were observed by a researcher (VH). Aspects

Record

NICU environment



Delivery room



Review



Figure 1 Record: recording equipment in the delivery room and on the neonatal intensive care unit (NICU): (1) eye-tracking glasses used for recording interventions performed on the NICU providing a point-of-view recording; (2) microphone; (3) camera, recording the hands of the providers during neonatal stabilisation; (4) camera, recording room-view during neonatal stabilisation. Review: during *Neoflix* sessions, medical and nursing staff evaluate a video.

of the sessions with regard to the safe learning environment, privacy, technical aspects, logistics and organisational aspects were reported in field notes, in consistence with the standards in qualitative research. After each VR session, all providers who participated in the session were

also asked to complete an anonymous questionnaire via a QR code and time was taken to let them fill it in. This questionnaire consisted of five questions using a Likert Scale (1—‘strongly disagree’, to 5—‘strongly agree’) questioning participants about their experience of the

review session and three open questions with a free text comment area available for remarks or comments (online supplemental file 2).

The observations regarding the implementation of VR and the feedback from the questionnaire from that session were interpreted by the *Neoflix* study group. The study group evaluated what could be improved about the guideline for VR. Action research was used to implement adjustments or expansions. These were tested and re-evaluated during following VR sessions. The results of the adjustments made through each improvement cycle to the guideline were noted and analysed by VH and RW, using content analysis to quantify concepts, themes and keywords.³⁰ The qualitative data analysis software *Atlas.Ti* (V.23) was used for coding. In consensus meetings, the study group identified the potential success factors and preconditions for successful VR that were included in the adjusted guideline. Using the framework of Rogers' diffusion of innovation theory,³¹ the success factors and preconditions were divided into different factors that influenced the rate of adoption of VR: (1) complexity (the degree to which VR as a tool is perceived as difficult to organise and use), (2) trialability (the degree to which VR can be tried for use by providers), (3) observability (the degree to which the results of VR can be observed and used in practice), (4) compatibility (the degree to which using VR for quality improvement is perceived as similar to beliefs and values of the neonatal care providers) and (5) relative advantage (the degree to which providers perceive how VR improves their current practice).

Participants' responses to the questionnaire were analysed qualitatively using conventional content analysis.³⁰ Quantitative data were analysed using IBM SPSS Statistics V.25 (IBM Software, Chicago, Illinois, 2016).

RESULTS

The redefined guideline is outlined in [figure 2](#).

Measurements during implementation of VR

A total of 18 sessions on neonatal stabilisation (6), endotracheal intubation or minimally invasive surfactant therapy (MIST) (6) and sterile line insertion (6) could be organised in the 9-month period. Sessions were taken without any major objections and with a total attendance of 297 providers. The mean (SD) attendance of each session was 17 (5) providers per session, including 7 (2) medical staff members, 8 (3) nursing staff members and 1 (1) medical student or researcher (other) ([figure 3A](#)), demonstrating that, in comparison with the baseline indicator, more members of the nursing staff attended the sessions but less medical staff members. A total of 47 neonatal procedures were recorded, including 23 recordings neonatal stabilisation, 8 recordings on intubation or surfactant administration and 8 recordings on sterile line insertion ([figure 3B](#)). All recordings were debriefed with the providers who were recorded. No one objected to the use of their recording for VR. At the end of the study period,

101/125 (81%) of all neonatal providers working on the NICU attended at least 1 session and 80/125 (64%) of all neonatal providers recorded their performance of a neonatal procedure at least 1 time ([figure 3C](#)).

In total, 179/297 (61%) of the providers attending VR completed the questionnaire ([figure 3D](#)); 72/179 (40%) by medical staff members, 104/179 (58%) by nursing staff members and 3/179 (2%) by medical students or researchers. Overall, the sessions were perceived as positive; 177/179 (99%) had a positive or strong positive opinion of the review sessions. Of all respondents, 169/179 (94%) agreed or strongly agreed that they had enough opportunity to speak during the VR sessions. Most, 173/179 (97%) of the providers who completed the questionnaire agreed or strongly agreed that they felt safe enough during the sessions to voice their opinion. Overall, 161/179 (90%) of the providers agreed or strongly agreed that VR sessions contributed to improvement in quality of care. The majority of the providers 170/179 (95%) agreed or strongly agreed that the 'feedback' rules were sufficiently used during the VR sessions. Qualitative feedback on participants' opinion included that the safe learning environment was a positive aspect of VR (online supplemental file 3). Sound quality and organisational aspects, such as having a closing round, were noted as areas for improvement. The most perceived barrier to speak for participants was lack of experience.

Redefining the guideline for VR

During the study period, in total 68 observations and/or considerations were noted and used as input for the improvement of the guideline, of which 38/68 (56%) were noted during the first 3 months of the project. Of these observations, 24/68 (35%) regarded the safe learning environment, 18/68 (27%) logistics, 10/68 (15%) how to organise VR in a multidisciplinary way, 10/68 (15%) technical aspects and 6/68 (9%) how to use VR for quality improvement.

Complexity

Providers had trouble remembering to use recording equipment. To make it less complex for providers, quick start manuals were developed. Reminders to record interventions were given on a regular basis at morning hand-overs, while voluntariness was always emphasised. It took about 2–3 hours for the chair to prepare a session, including preview with all providers involved. To distribute the workload, different staff members would alternate chairing the sessions. To make sure it could fit in the busy time schedules of providers, it was important to stay within the 30 min time limit planned for a session, including discussion. The *Neoflix* team decided in advance if the sessions would focus on certain aspects of a video or if the discussion was to flow freely, based on the intensity of provided care. After the first few sessions, it was noted that sound quality was not sufficient for evaluating communication. To reduce the background noise in the delivery room, the *Rode wireless Go* (Rode, Sydney,

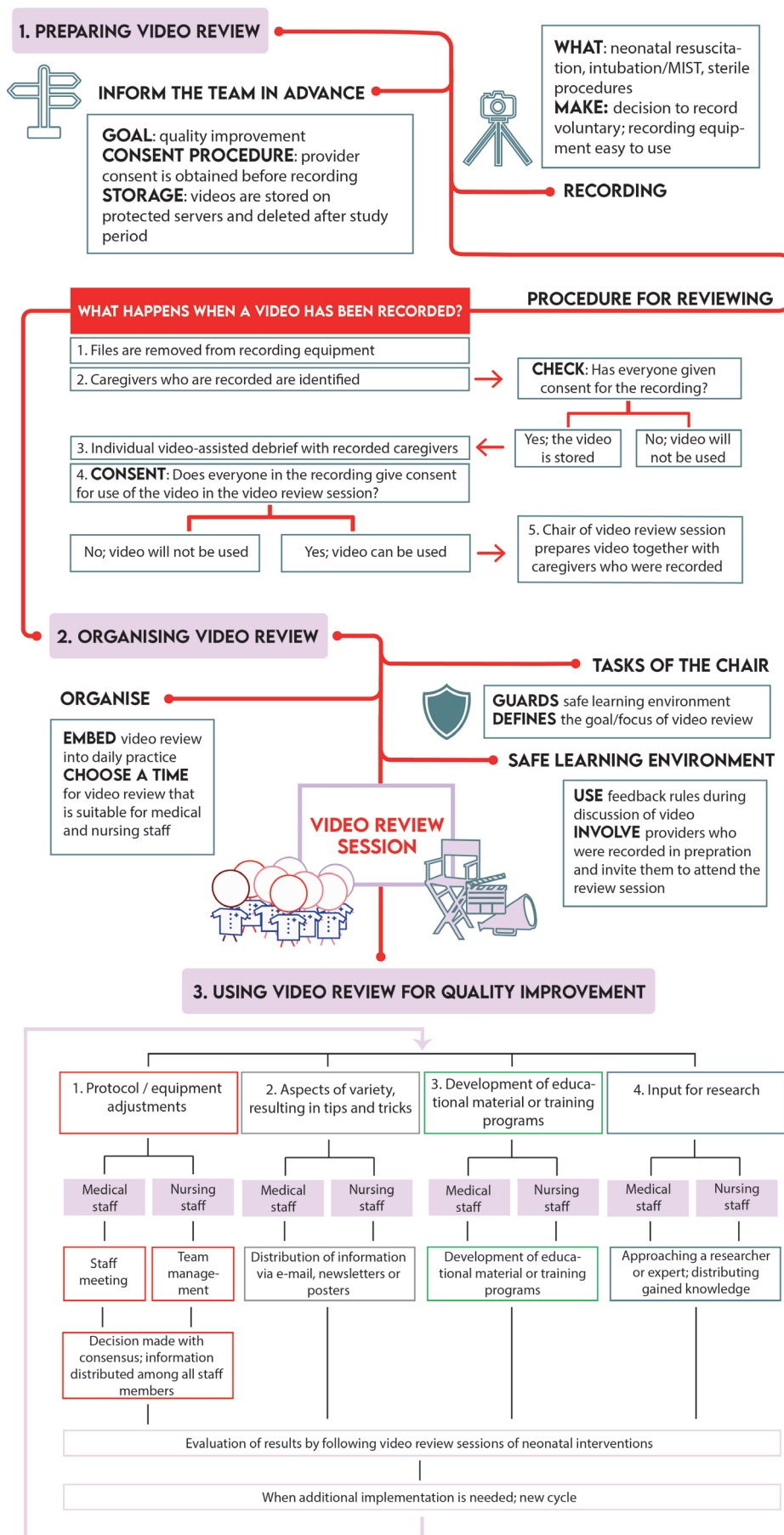


Figure 2 (Re)defined guideline for implementing video review of neonatal procedures.

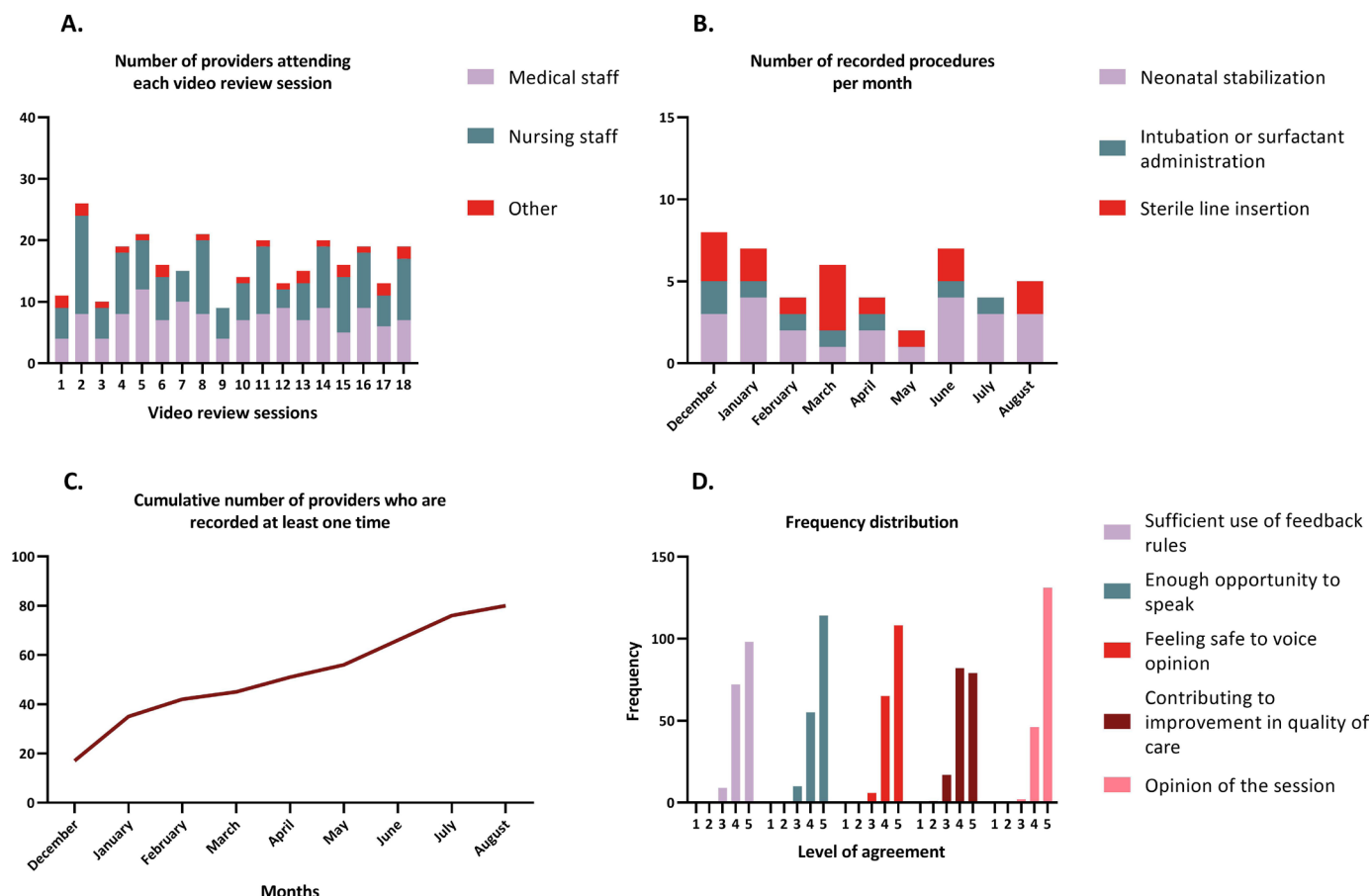


Figure 3 Outcome measurements during the implementation of video review. (A) Number of providers attending each video review session. (B) Number of recorded procedures per month. (C) Cumulative number of providers who are recorded at least one time. (D) Frequency distribution. The scale for the level of agreement consists of 1=strongly disagree, 2=disagree, 3=neither agree nor disagree, 4=agree, 5=strongly agree. Except for *opinion of the sessions*, which consists of 1=strong negative, 2=negative, 3=neutral, 4=positive and 5=strong positive.

Australia) microphones were added for improved sound recording.

Trialability

Before implementation, the team was informed about the goal and ‘rules’ of VR. During implementation, the necessity of defining the pathway for obtaining consent and preparation of VR was identified (figure 2). Using this pathway, providers knew that a recording would not be used for VR without their consent. No providers declined use of the video for VR during the study period. No medicolegal issues occurred during the study project. In addition, videos of neonatal stabilisation at birth were reviewed with parents and consent to use the videos in the review session was obtained in all cases.

Observability

VR resulted in immediate tips and tricks that could be used by providers in practice. By introducing a closing round to end the session, the lessons learnt were repeated and it was determined what could be concluded from the session. If a session resulted in an adaptation of protocol/equipment or changes in behaviour, this effect could be checked in next review sessions. Additionally, a newsletter

was sent out with lessons learnt from VR. Hereby, the results of VR were visible immediately. The meeting was set up in a hybrid manner, allowing for physical and online attendance, and online participants were actively involved. The meeting was scheduled in the afternoon at a time when both nursing staff and medical staff could attend, leading to a balanced distribution of disciplines between participants and dynamic discussions where multiple viewpoints could be considered. Especially the combination of the nursing and medical staff was seen as valuable, as there are few moments on a NICU where medical and nursing staff openly discuss together in detail how they wish to improve care.

Compatibility

An important precondition of VR was having a safe learning environment during the sessions. Different members of the nursing and the medical staff were part of the study group and were trained to chair sessions. The chair had to be able to guide the discussion and had to have a position of trust within the NICU team. Recorded providers should not feel any apprehension when sharing their perspective on the recorded procedure. The viewing

of a video with the participants prior to the session gave the necessary context and was considered a success factor and an important precondition for a safe learning environment. Three rules were used to ensure feedback was given constructively and respectfully during the sessions: (1) be specific and to the point, (2) describe the situation without judgement and (3) to focus on behaviour and do not make comments personal. The dedicated chair oversaw that participants adhered to these rules and were respectful when commenting on the video. Also, at the start of the session, it was emphasised that the aim of VR was not to assess someone's performance or to strive for perfection, but to learn from how the intervention is performed in the specific context and setting.

Relative advantage

The input from providers attending the session was meant to be used as the basis for quality improvement of the interventions. To enable this, structured pathways were developed by the study group (figure 2). After each session, the study group evaluated the discussion during VR and determined the nature of the lessons learnt or insights from the session. The items could then be divided into four categories with each their own implementation strategy: (1) findings used to propose an adjustment in protocols or equipment; (2) tips and tricks, for example, aspects of variety or best practices; (3) development of training programmes or educational material; and (4) findings that needed research to clarify. Developing these structured pathways allowed every point or insight that came out of the sessions to be used to drive change and to be communicated to the team.

LESSONS AND LIMITATIONS

Although VR has proven to be beneficial for guideline compliance and clinical performance of emergency care teams,^{7 9–12 17 32} studies have not focused on resolving challenges that influence implementation.^{19 29} This study describes a novel way of using VR as an initiative to improve neonatal care by recording three different procedures and discussing these in interprofessional VR sessions, focused on having a safe learning environment. Recognising the dynamic and contextual nature of VR, we chose action research over the Plan, Do, Study, Act methodology to make implementation provider driven and empowering.³³ Through this approach, the study group, informed by both observations and provider feedback, planned and implemented actions, and evaluated their effectiveness within the context of VR. As it is difficult to measure the effect of individual adjustments, the study group used the Diffusion of Innovation model to identify key characteristics influencing VR adoption and defined the preconditions and success factors for its ongoing implementation.^{31 34} Our results show that implementation was successful with a total of 18 sessions that could be organised and completed without any major objections and with a consistent attendance of providers. There were on average 17 providers

attending each session, with an equal distribution of medical and nursing staff. Also, different providers recorded neonatal procedures continuously during the study period. The healthcare providers were positive about VR in our department and felt that there was a safe learning environment present during the sessions, which could be attributed to our improved guideline for implementing VR. Based on our findings, we would recommend using an action research approach for implementing VR as it is important to include the stakeholders in the organisation. Also, following the observed preconditions and success factors, we recommend to focus implementation of VR on keeping it safe, small and simple.

Previous studies recommend informing the participants on the purpose of VR, to create a sense of urgency and motivation.^{35 36} We also observed that creating enthusiasm as well as a sense of safety among the providers is important. In concurrence with previous studies, it is important to be fully transparent about the project and consent procedures.^{20 21 27} The pathway for obtaining consent and preparing the VR session was defined to enable providers to have the final say in whether a video would be used for VR. Although no one objected to the use of their recordings for the review sessions, we are not sure whether participants felt duty-bound. Further studies are needed to investigate this.

As VR has been used in several domains of healthcare, for example, patient care, quality improvement, research and education, its use is also complicated by conflicting issues and requirements in each of these domains.³⁷ The different stakeholders and ethical values surrounding VR could withhold other centres from implementing it. Although it requires careful consideration of the different benefits and disadvantages to the stakeholders, we hope our approach offers some guidance in setting up VR. The goal of VR on our department was to improve the practice of neonatal care and the non-identifiable recordings were already part of standard care; therefore, no patient consent was asked for these recordings. However, if parents were visible in the identifiable recordings, they were asked for consent for protection of privacy, but were also offered the possibility of reviewing the recording. Previous research showed that parents appreciated this openness and transparency from providers, making them less likely to hold providers responsible for medical errors.³⁸ Providers always needed to consent for the identifiable recordings in order to protect their privacy and to make recording voluntarily. Because VR was meant for quality assurance purposes, identifiable recordings were deleted at the end of the project, potentially limiting the use of VR for further research. Of note, if an adverse event had occurred during an intervention, this had to be reported regardless of whether a video had been made or not. Although recordings used for quality improvement are often protected by law, it is not clear if they can be used as evidence in court.³⁷ Medicolegal problems with video recording have so far not been reported,^{28 39} nor did we encounter these.

We did not record every intervention as standard procedure and for this reason, a selection bias in the recorded procedures could have occurred. However, the goal of VR was to start the conversation between the members of the medical and the nursing staff about how they wished to improve procedures together. It was more about sharing different points of view on clinical care than about recording and assessing every procedure standardly. Nevertheless, we made sure that the barrier to record a procedure with recording equipment was as low as possible. Although more cameras present to record the procedures would provide a more comprehensive view, it would also take more time to set up. Therefore, we chose one or two camera angles to record the neonatal procedures to minimise the time required to prepare for recording. During implementation, we observed that the safe learning environment was an ongoing learning process. This can be influenced by team dynamics, the team's experience, but also by the set-up of the meetings with effective 'feedback' rules. Den Boer *et al* already noticed that previous VR meetings on our department evolved into more open discussions when VR sessions were performed regularly.¹⁵ Assigning different dedicated chairs to guide the sessions and guarding the safe learning environment were considered a success factor. Putting emphasis on including nursing staff was seen as a key factor in achieving valuable discussions during VR.

VR is now embedded in our department as an educational tool and for quality improvement. The structured pathways and corresponding strategies added to the guideline were useful to implement lessons learnt from VR for improving quality. Therefore, the staff members who were not present during VR would know what insights or lessons were learnt from the sessions and how they would be used to drive change. We currently did not aim for measuring the effect of VR on improving care, but we will use our improved guideline and perform studies with the focus on the value of VR for improving neonatal procedures.

The study group who evaluated the VR sessions consisted of different stakeholders, who were supposed to represent the different perspectives involved in neonatal care. They influenced how improvement cycles were used to change the set-up and organisation of VR. To include all providers and give them equal voices in this process, the anonymous questionnaire was added, which gave all providers equal opportunity to provide input on how they felt experienced VR. Our questionnaire did not evaluate clinical outcomes but self-reported perceptions of the VR sessions. However, it did show that VR was well received and ongoing studies on our department will focus on the effect of interprofessional VR on quality of care and clinical outcomes.

We report an implementation in a single institution where providers were already familiar with VR. This might have influenced the success of the implementation and limits the generalisability of our results. However, the use of identifiable videos with audio and discussing this

with members of the medical and nursing staff together was new to all providers.

CONCLUSION

We successfully implemented a provider-accepted VR program within our NICU using an action research approach, marking the first steps for continuous evaluation of neonatal care through interprofessional VR. The project fostered a dedicated study group who refined VR guidelines, resulting in widespread support from the NICU team. To ensure program sustainability and a safe learning environment in the future, the *Neoflix* group will continue driving project goals and facilitating VR sessions using the redefined guidelines. This optimised framework is expected to diminish time and resource constraints. However, ongoing assessment of the VR sessions will be crucial for both program longevity and a secure learning atmosphere. By sharing our experience with guideline adaptation, we empower other centres to tailor this framework to their specific needs. Our findings highlight the importance of securing team buy-in and establishing a dedicated VR team for successful program implementation.

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Competing interests None declared.

Patient and public involvement A member from the quality and patient safety department was involved with the design and conduct of the study. The goal of this study was to define a guideline for healthcare providers, so providers were actively involved in the action research cycles.

Patient consent for publication Not applicable.

Ethics approval This study was reviewed by the Ethics Review Committee of the LUMC (N21.169). In concordance with laws and guidelines, a statement of no objection against execution of the study was issued, exempted this study. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request.

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